MELTING AND BOILING POINTS

Name: ..... **Date:** .....

Below its melting point a substance is solid; between its melting and boiling points it is liquid; above its boiling point it is a gas. Part 1: complete the three state columns. Part 2: answer using the table.

PART 1: WHAT STATE IS IT?

Substance	Melting point (°C)	Boiling point (°C)	State at 20 °C	State at −50 °C	State at 500 °C
Oxygen	−219	−183			
Water	0	100			
Ethanol	−114	78			
Mercury	−39	357			
Iron	1538	2862			
Table salt (NaCl)	801	1465			
Nitrogen	−210	−196			
Gold	1064	2856			
Butane	−138	−1			
Aluminium	660	2519			

PART 2: USE THE DATA

Which substance has the lowest melting point?

Which substances are liquids at room temperature (20 °C)?

Which substance is a liquid over the widest range of temperatures? Show your working.

Mercury was used in thermometers for 300 years. Use the table to explain why.

A substance melts at −7 °C and boils at 59 °C. What state is it at 20 °C? (It is bromine.)

What is the difference between evaporation and boiling?

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MELTING AND BOILING POINTS ANSWER KEY

Name: ..... **Date:** .....

PART 1: WHAT STATE IS IT?

Oxygen | | | gas | gas | gas

Water | | | liquid | solid | gas

Ethanol | | | liquid | liquid | gas

Mercury | | | liquid | solid | gas

Iron | | | solid | solid | solid

Table salt | | | solid | solid | solid

Nitrogen | | | gas | gas | gas

Gold | | | solid | solid | solid

Butane | | | gas | liquid | gas

Aluminium | | | solid | solid | solid

PART 2: USE THE DATA

1. Oxygen (−219 °C) 2. Water, ethanol and mercury 3. Aluminium: 2519 − 660 = 1859 degrees (gold 1792, iron 1324, salt 664). 4. It is a liquid across the whole everyday range, from −39 °C to 357 °C, so it can expand up the tube without freezing or boiling. 5. Liquid: 20 °C is between the melting and boiling points. 6. Evaporation happens only at the surface and at any temperature below the boiling point; boiling happens throughout the liquid, with bubbles, at the boiling point.